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Exercise Sheet 6

This is not the official exercise sheet from the course.
More difficult exercises are denoted by ★ to ★★★

> If you can solve it, it is an exercise;
> otherwise it's a research problem.
>
> Richard Bellman

Exercise 1.
Compute the following remainders

1. $R_{11}(24^{100} \cdot 33^{50})$.
2. $R_7(8^{2025} + 14^{2026})$.
3. $R_5(12^{2026} \cdot 3^{100})$.
4. $R_9(18^{100} + 28^{50})$.

Solution 1.

1. Since $33 \equiv 0 \pmod{11}$,
$$24^{100} \cdot 33^{50} \equiv 24^{100} \cdot 0 \equiv 0 \pmod{11}.$$
So the remainder is 0.
2. Since $8 \equiv 1 \pmod{7}$ and $14 \equiv 0 \pmod{7}$,
$$8^{2025} + 14^{2026} \equiv 1^{2025} + 0^{2026} \equiv 1 \pmod{7}.$$
So the remainder is 1.
3. Modulo 5, $12 \equiv 2$ and $3 \equiv 3$. Since
$$2^2 \equiv -1 \pmod{5}, \qquad 2^4 \equiv 1 \pmod{5},$$
and $2026 = 4 \cdot 506 + 2$, we get
$$12^{2026} \equiv 2^{2026} \equiv 2^2 \equiv 4 \pmod{5}.$$
Also,
$$3^2 \equiv -1 \pmod{5}, \qquad 3^4 \equiv 1 \pmod{5}.$$
Since $100 = 4 \cdot 25$,
$$3^{100} \equiv 1 \pmod{5}.$$
Therefore
$$12^{2026} \cdot 3^{100} \equiv 4 \cdot 1 \equiv 4 \pmod{5}.$$
So the remainder is 4.

4. Since $18 \equiv 0 \pmod 9$ and $28 \equiv 1 \pmod 9$,

$$18^{100} + 28^{50} \equiv 0 + 1 \equiv 1 \pmod 9.$$

So the remainder is 1.

Exercise 2.
Prove that $13 \mid 2^{70} + 3^{70}$.

Solution 2.
First notice that we have

$$2^6 = 64 \equiv -1 \pmod{13}.$$

Thus

$$2^{70} = 2^{6 \cdot 11 + 4} = (2^6)^{11} 2^4 \equiv (-1)^{11} \cdot 16 \equiv -16 \equiv -3 \pmod{13}.$$

Also,

$$3^3 = 27 \equiv 1 \pmod{13}.$$

Since $70 = 3 \cdot 23 + 1$,

$$3^{70} \equiv 3 \pmod{13}.$$

Therefore

$$2^{70} + 3^{70} \equiv -3 + 3 \equiv 0 \pmod{13},$$

so

$$13 \mid 2^{70} + 3^{70}.$$

Exercise 3.
Let $n \in \mathbb{N} \setminus \{0\}$. Prove that $n^2 \mid (n+1)^n - 1$.

Solution 3.
By the binomial theorem,

$$(n+1)^n = (1+n)^n = 1 + \binom{n}{1} n + \binom{n}{2} n^2 + \cdots + \binom{n}{n} n^n.$$

Since $\binom{n}{1} = n$,

$$(n+1)^n = 1 + n^2 + \binom{n}{2} n^2 + \binom{n}{3} n^3 + \cdots + \binom{n}{n} n^n.$$

Every term after the first is divisible by n^2. Hence

$$(n+1)^n - 1$$

is divisible by n^2, so

$$n^2 \mid (n+1)^n - 1.$$

Exercise 4.
Let $k \in \mathbb{N}$. Prove that $19 \mid 2^{2^{6k+2}} + 3$.

Solution 4.
First for $k = 0$, have

$$2^{2^2} + 3 = 19$$

which is clearly divisible by 19. So assume that $k \geq 1$.

Notice that we have

$$2^9 = 512 \equiv -1 \pmod{19},$$

so

$$2^{18} \equiv 1 \pmod{19}.$$

Now

$$2^{6k+2} = 4 \cdot 64^k.$$

Since we have

$$64 \equiv 10 \pmod{18} \quad \text{and} \quad 10^2 \equiv 100 \equiv 10 \pmod{18}$$

So for $k \geq 1$,

$$64^k \equiv 10^k \equiv 10 \pmod{18}.$$

Thus

$$2^{6k+2} = 4 \cdot 64^k \equiv 4 \cdot 10 = 40 \equiv 4 \pmod{18}.$$

So

$$2^{6k+2} = 18q + 4$$

for some integer q. Therefore

$$2^{2^{6k+2}} = (2^{18})^q 2^4 \equiv 1^q \cdot 16 \equiv 16 \equiv -3 \pmod{19}.$$

Hence

$$2^{2^{6k+2}} + 3 \equiv -3 + 3 \equiv 0 \pmod{19},$$

so

$$19 \mid 2^{2^{6k+2}} + 3.$$

Exercise 5.
Let $a, b \in \mathbb{Z}$ with $b > 0$. Suppose a can be expressed in the following ways using the division algorithm

$$a = bq_1 + r_1 \quad a = bq_2 + r_2$$

for $0 \leq r_1 < b$ and $0 \leq r_2 < b$.

Prove that $q_1 = q_2$ and $r_1 = r_2$.

Solution 5.
Suppose

$$a = bq_1 + r_1 = bq_2 + r_2,$$

where $b > 0$ and

$$0 \leq r_1 < b, \qquad 0 \leq r_2 < b.$$

Subtracting gives

$$b(q_1 - q_2) = r_2 - r_1.$$

Thus $b \mid r_2 - r_1$. But since $0 \leq r_1 < b$ and $0 \leq r_2 < b$,

$$-b < r_2 - r_1 < b.$$

The only multiple of b in this range is 0. Hence

$$r_2 - r_1 = 0,$$

so $r_1 = r_2$. Then

$$b(q_1 - q_2) = 0,$$

and since $b > 0$,

$$q_1 = q_2.$$

Therefore the quotient and remainder in the division algorithm are unique.

Exercise 6.
Find all solutions to the following system of congruences

$$\begin{aligned} 3x + 4y &\equiv 5 \pmod{13} \\ 2x + 5y &\equiv 7 \pmod{13} \end{aligned}$$

Solution 6.
We solve

$$\begin{cases} 3x + 4y \equiv 5 \pmod{13}, \\ 2x + 5y \equiv 7 \pmod{13}. \end{cases}$$

Multiply the first congruence by 2:

$$6x + 8y \equiv 10 \pmod{13}.$$

Multiply the second congruence by 3:

$$6x + 15y \equiv 21 \equiv 8 \pmod{13}.$$

Subtracting the first from the second gives

$$7y \equiv 8 - 10 \equiv -2 \equiv 11 \pmod{13}.$$

Since $7 \cdot 2 = 14 \equiv 1 \pmod{13}$, multiply by 2:

$$y \equiv 2 \cdot 11 = 22 \equiv 9 \pmod{13}.$$

Substitute $y \equiv 9$ into the first congruence:

$$3x + 4 \cdot 9 \equiv 5 \pmod{13}.$$

Since $36 \equiv 10 \pmod{13}$,

$$3x + 10 \equiv 5 \pmod{13},$$

so

$$3x \equiv -5 \equiv 8 \pmod{13}.$$

Since $3 \cdot 9 = 27 \equiv 1 \pmod{13}$, multiply by 9:

$$x \equiv 9 \cdot 8 = 72 \equiv 7 \pmod{13}.$$

Thus all solutions are

$$x \equiv 7 \pmod{13}, \qquad y \equiv 9 \pmod{13}.$$

Equivalently,

$$x = 7 + 13m, \qquad y = 9 + 13n, \qquad m, n \in \mathbb{Z}.$$

Exercise 7.
Prove that there are no $x, y \in \mathbb{Z}$ such that

$$x^3 - 3y^3 = 9$$

Solution 7.
Assume, for contradiction, that there exist integers x, y such that

$$x^3 - 3y^3 = 9.$$

Reduce modulo 9. We have that $x^3 \bmod 9$ only takes the following values

$$0,\ 1,\ -1.$$

Also $3y^3$ modulo 9 can only be

$$0,\ 3,\ 6.$$

Since the right-hand side is $9 \equiv 0 \pmod{9}$, we need

$$x^3 \equiv 3y^3 \pmod{9}.$$

But $3y^3$ is never congruent to 1 or -1 modulo 9, so necessarily

$$x^3 \equiv 0 \pmod{9}.$$

Hence $3 \mid x$. Write $x = 3u$. Then

$$27u^3 - 3y^3 = 9.$$

Dividing by 3,

$$9u^3 - y^3 = 3.$$

Modulo 3, this gives

$$-y^3 \equiv 0 \pmod{3},$$

so $3 \mid y$. Write $y = 3v$. Then the original equation becomes

$$(3u)^3 - 3(3v)^3 = 9,$$

so

$$27u^3 - 81v^3 = 9.$$

The left-hand side is divisible by 27, but 9 is not divisible by 27. Contradiction.

Therefore there are no integers x, y such that

$$x^3 - 3y^3 = 9.$$

Exercise 8.
Prove that there are no $x, y \in \mathbb{Z}$ such that

$$x^4 - 5y^2 = 2$$

Solution 8.
Assume, for contradiction, that there exist integers x, y such that

$$x^4 - 5y^2 = 2.$$

Work modulo 4. For any integer x,

$$x^4 \equiv 0 \pmod 4 \quad \text{if } x \text{ is even,}$$

and

$$x^4 \equiv 1 \pmod 4 \quad \text{if } x \text{ is odd.}$$

Also, for any integer y,

$$y^2 \equiv 0 \text{ or } 1 \pmod 4.$$

Since $5 \equiv 1 \pmod 4$,

$$5y^2 \equiv y^2 \pmod 4.$$

Thus

$$x^4 - 5y^2 \equiv x^4 - y^2 \pmod 4.$$

The possible values of $x^4 - y^2 \pmod 4$ are

$$0,\ 1,\ 0 - 1 \equiv 3,\ 1 - 1 = 0.$$

So

$$x^4 - 5y^2 \equiv 0, 1, \text{ or } 3 \pmod 4,$$

never 2. But the right-hand side is $2 \equiv 2 \pmod 4$. Contradiction.

Therefore there are no integers x, y such that

$$x^4 - 5y^2 = 2.$$

★ **Exercise 9.**
Prove that there are no $x, y \in \mathbb{Z}$ such that

$$x^3 - 3 = y^2$$

Hint: For a prime p such that $p \nmid a$ we have $a^{p-1} \equiv 1 \pmod{p}$

Solution 9.
Assume for the sake of contradiction that there exist integers x and y satisfying:

$$x^3 - 3 = y^2 \implies x^3 = y^2 + 3$$

Suppose x is even. Then we get that $x^3 \equiv (2k)^3 \equiv 0 \pmod{8}$. Substituting gives

$$y^2 \equiv -3 \equiv 5 \pmod{8}$$

Now we compute $y^2 \pmod{8}$ for all possible values Since $y^2 \not\equiv 5 \pmod{8}$ for any y, we conclude

y	0	1	2	3	4	5	6	7
y^2	0	1	4	1	0	1	4	1

that x must be odd.

Now we consider the equation mod 4 and mod 9.

Since x is odd we get that $x \equiv 1 \pmod{4}$ or $x \equiv 3 \pmod{4}$.

If $x \equiv 1 \pmod{4}$, then $x^3 \equiv 1 \pmod{4}$, so

$$y^2 = x^3 - 3 \equiv -2 \equiv 2 \pmod{4}$$

Since square can only be 0 or 1 mod 4, this is impossible. Thus we must have $x \equiv 3 \pmod{4}$.

Now we compute $x^2, x^3 \pmod{9}$

x	0	1	2	3	4	5	6	7	8
x^2	0	1	4	0	7	7	0	4	1
x^3	0	1	8	0	1	8	0	1	8

The possible quadratic residues modulo 9 are $\{0, 1, 4, 7\}$. We check the candidate values for $y^2 = x^3 - 3 \pmod{9}$ based on $x^3 \pmod{9}$:

- If $x^3 \equiv 0 \pmod{9} \implies y^2 \equiv 0 - 3 \equiv 6 \pmod{9}$
- If $x^3 \equiv 8 \pmod{9} \implies y^2 \equiv 8 - 3 \equiv 5 \pmod{9}$
- If $x^3 \equiv 1 \pmod{9} \implies y^2 \equiv 1 - 3 \equiv 7 \pmod{9}$

Since only $7 \in R_9(x^2)$, we must have that $x^3 \equiv 1 \pmod 9$. We also get that $x^3 \equiv 1 \pmod 4$ if and only if $x \in \{1, 4, 7\}$, which gives that we must have

$$x \equiv 1 \pmod 3$$

So to summarize, we know that it must hold that

$$x \equiv 1 \pmod 3 \quad \text{and} x \equiv 3 \pmod 4$$

Now we add 4 to both sides of our original equation to factor it

$$y^2 + 4 = x^3 + 1$$
$$y^2 + 4 = (x + 1)(x^2 - x + 1)$$

Since x is odd, y must also be even ($y = 2k$). Substituting $y = 2k$ gives

$$4(k^2 + 1) = (x + 1)(x^2 - x + 1)$$

Now consider the factor $N = x^2 - x + 1$. Since $x \equiv 3 \pmod 4$

$$N = x^2 - x + 1 \equiv 3^2 - 3 + 1 \equiv 7 \equiv 3 \pmod 4$$

Any integer congruent to $3 \pmod 4$ must contain at least one prime factor $p \equiv 3 \pmod 4$.

Since $p \mid N$, it follows that $p \mid 4(k^2 + 1)$. Because p is an odd prime ($p \neq 2$), $\gcd(p, 4) = 1$, which forces

$$p \mid (k^2 + 1) \implies k^2 \equiv -1 \pmod p$$

Raising both sides to $\frac{p-1}{2}$ (an odd integer since $p \equiv 3 \pmod 4$)

$$(k^2)^{\frac{p-1}{2}} \equiv (-1)^{\frac{p-1}{2}} \implies k^{p-1} \equiv -1 \pmod p$$

By Fermat's Little Theorem, $k^{p-1} \equiv 1 \pmod p$, so $1 \equiv -1 \pmod p \implies p \mid 2$, contradicting $p \equiv 3 \pmod 4$.

Therefore, no integer solutions exist.

★ **Exercise 10.**
Prove that $\gcd(n, 2^{2^n} + 1) = 1$ for $n = 1, 2, \ldots$

Solution 10.
Assume, for the sake of contradiction, that there exists a positive integer n such that $\gcd(n, 2^{2^n} + 1) > 1$. Then there exists a prime p such that

$$p \mid n \quad \text{and} \quad p \mid \left(2^{2^n} + 1\right).$$

Since $2^{2^n} + 1$ is odd, p must be an odd prime ($p \neq 2$).

From $p \mid \left(2^{2^n} + 1\right)$, we have

$$2^{2^n} \equiv -1 \pmod{p}.$$

Squaring both sides yields

$$2^{2^{n+1}} \equiv 1 \pmod{p}.$$

By Fermat's Little Theorem, since p is an odd prime and $\gcd(2, p) = 1$, we also have

$$2^{p-1} \equiv 1 \pmod{p}.$$

Let $d = \gcd(p - 1, 2^{n+1})$. By Bézout's Identity (Lemma 4.5), there exist integers $a, b \in \mathbb{Z}$ such that $d = a(p - 1) + b(2^{n+1})$. Therefore

$$2^d = 2^{a(p-1)+b(2^{n+1})} = \left(2^{p-1}\right)^a \cdot \left(2^{2^{n+1}}\right)^b \equiv 1^a \cdot 1^b \equiv 1 \pmod{p}.$$

Since d divides 2^{n+1}, d must be a power of 2, so $d = 2^k$ for some integer k with $0 \leq k \leq n + 1$.

We claim that $k = n + 1$. If $k \leq n$, then $d = 2^k$ divides 2^n. Writing $2^n = m \cdot d$ for some integer $m \geq 1$, we would have

$$2^{2^n} = (2^d)^m \equiv 1^m \equiv 1 \pmod{p}.$$

However, this contradicts $2^{2^n} \equiv -1 \pmod{p}$, because $1 \equiv -1 \pmod{p}$ implies $p \mid 2$, which is impossible as p is an odd prime.

Thus, we must have $k = n + 1$, which means $d = 2^{n+1}$. Since $d \mid (p - 1)$, it follows that $2^{n+1} \mid (p - 1)$, and consequently

$$p \geq 2^{n+1} + 1.$$

On the other hand, since $p \mid n$, we have $p \leq n$. Combining these inequalities yields

$$2^{n+1} + 1 \leq p \leq n,$$

which is impossible because $n < 2^{n+1} + 1$ for all positive integers n.

By contradiction, no such prime p exists, and therefore $\gcd(n, 2^{2^n} + 1) = 1$.